Exercises on Streaming

Exercise 1 *With reference to the Boyer-Moore algorithm which finds the majority in a stream Σ of n elements $x_1, x_2, \ldots, x_n$, if one exists, show that the following invariant is maintained by the algorithm. For every $t \geq 0$, after processing x_t (if $t = 0$ this is the start of the algorithm, before any element is processed)* :

- *There are* count *occurrences of* cand *in $x_1, x_2, \ldots, x_t$.*
- *Besides the* count *occurences of* cand*, the remaining $t-$count elements can be grouped into $(t - \text{count})/2$ pairs (e_1, e_2) with $e_1 \neq e_2$.*

Solution. For $t = 0$, the invariant trivially holds since count $= 0$. Suppose that it holds after processing $x_1, x_2, \ldots, x_{t-1}$, for some $t - 1 \geq 0$, and let count $= c$ and cand $= e$ at that time. To show that it holds after processing x_t, we distinguish among three cases:

- **CASE 1:** $c = 0$. In this case, after processing x_t we have count $= 1$ and cand $= x_t$. Hence, we have 1 occurence of cand in $x_1, x_2, \ldots, x_t$, and, from the invariant assumed to be true after processing x_{t-1}, we see that the elements $x_1, x_2, \ldots, x_{t-1}$ can be grouped into $(t - 1 - c)/2 = (t - 1)/2$ pairs (e_1, e_2) with $e_1 \neq e_2$.
- **CASE 2:** $c > 0$ and $x_t = e$. In this case, after processing x_t we have count $= c + 1$ and cand $= e = x_t$. Hence, considering also x_t we have $c + 1$ occurences of cand in $x_1, x_2, \ldots, x_t$, and, from the invariant assumed to be true after processing x_{t-1}, we see that the remaining elements can be grouped into $(t - 1 - c)/2 = (t - (c + 1))/2$ pairs (e_1, e_2) with $e_1 \neq e_2$.
- **CASE 3:** $c > 0$ and $x_t \neq e$. In this case, after processing x_t we have count $= c-1$ and cand $= e$. From the invariant assumed to be true after processing x_{t-1}, we obtain that there are $c - 1$ occurences of cand in $x_1, x_2, \ldots, x_t$, and that the remaining elements can be grouped into $1 + (t - 1 - c)/2 = (t - (c - 1))/2$ pairs (e_1, e_2) with $e_1 \neq e_2$, where the additional pair (cand, x_t) has been added to them.

□

Exercise 2 *Let $\Sigma = x_1, x_2, \ldots x_n$ be a stream of n distinct integers in $[1, n + 1]$. Design a streaming algorithm that finds the missing number using $O(1)$-size working memory, 1 pass, and $O(1)$ update and query time.*

Solution. Assume that n is known to the algorithm and observe that $\sum_{a=1}^{n+1} a = (n + 1)(n + 2)/2$. Then it is sufficient to initialize a counter to 0, and to add each x_i to it upon arrival. It is immediate to see that after processing x_n, then $(n+1)(n+2)/2-$counter gives the missing number. The algorithm clearly requires $O(1)$ working memory, 1 pass, and

$O(1)$ update and query time. If n is unknown, one can simply maintain another variable that counts the number of elements processed. Also, the approach can be extended to the case when there are 2 missing numbers: it is sufficient to exploit the well-known properties $\sum_{a=1}^{\ell} a = \ell(\ell+1)/2$ and $\sum_{a=1}^{\ell} a^2 = \ell(\ell+1)(2\ell+1)/6$. □

Exercise 3 *Let $\Sigma = x_1, x_2, \ldots x_n$ be a stream of n items and let S be an m-sample computed on Σ using reservoir sampling. For a given frequency threshold $\phi \in (0,1)$, consider a frequent item a, that is an item occurring at least ϕn times in Σ. Show that, if $m \geq 1/\phi$, at least one occurrence of a is in S, in expectation.*

Solution. Let m_a be the number of occurrences of a in S. (Note that reservoir sampling does not check the identity of the items, so it may add several distinct occurrences of the same item to the sample.) Now, m_a is a random variable and we need to show that $\mathrm{E}[m_a] \geq 1$. Suppose that a occurs k times in Σ, hence $k \geq \phi n$, and let x_{i_j} be the j-th occurrence of a in Σ, with $1 \leq j \leq k$ and $1 \leq i_1 < i_2 < \cdots < i_k \leq n$. Define the following k Bernoulli variables:

$$Y_j = \begin{cases} 1 & \text{if } x_{i_j} \in S \\ 0 & \text{else,} \end{cases}$$

for $1 \leq j \leq k$. By the property of reservoir sampling, we have that $\Pr(Y_j = 1) = m/n$. Since $m_a = \sum_{1 \leq j \leq k} Y_j$, we have that by linearity of expectation

$$\mathrm{E}[m_a] = \mathrm{E}\left[\sum_{1 \leq j \leq k} Y_j\right] = \sum_{1 \leq j \leq k} \mathrm{E}[Y_j] = km/n \geq \phi m \geq 1,$$

where the last inequality follows from the assumption $m \geq 1/\phi$. □

Exercise 4 *Suppose that S_1 and S_2 are two m-samples extracted from two streams Σ_1 and Σ_2 of size n each. Given S_1 and S_2, show how to construct an m-sample of the concatenation of the two steams $\Sigma_1 \cdot \Sigma_2$ (assume that all elements of the two streams are distinct).*

Solution. An m-sample of $\Sigma_1 \cdot \Sigma_2$ is a subset S of exactly m elements from $\Sigma_1 \cdot \Sigma_2$ such that every $x \in \Sigma_1 \cdot \Sigma_2$ has probability $m/(2n)$ to be in S. Given S_1 and S_2, the desired m-sample can be obtained by computing an m-sample S of $S_1 \cup S_2$. Indeed, for each element $x \in \Sigma_1$, we have:

$$\Pr[x \in S] = \Pr[x \in S_1] \cdot \Pr[x \in S | x \in S_1] = (m/n) \cdot (m/(2m)) = m/2n.$$

A similar argument holds for each $x \in \Sigma_2$. Therefore S is an m-sample of $\Sigma_1 \cdot \Sigma_2$. □

Exercise 5 *Let $S = \{s_1, s_2, \ldots, s_m\}$ be an m-sample extracted from a stream Σ of n items labeled either* red *or* blue*. Let $R \in [1,n]$ be the number of red items in Σ and let X_S be the number of red items in S. Show that $(n/m) \cdot X_S$ is an unbiased estimator of R, that is that $\mathrm{E}[(n/m) \cdot X_S] = R$.*

Solution. Let $a_1, a_2, \ldots, a_R$ be the red items in Σ, and, for $1 \le i \le R$, define the Bernoulli random variable

$$X_{a_i} = \begin{cases} 1 & \text{if } a_i \in S \\ 0 & \text{else} \end{cases}$$

Clearly, $X_S = \sum_{i=1}^{R} X_{a_i}$. For any i, the expectation of any X_{a_i} is equal to $\Pr(X_{a_i} = 1)$ which, by definition of m-sample, is m/n. Therefore, using the linearity of expectation, we have that

$$\mathrm{E}[(n/m)\cdot X_S] = (n/m)\cdot \mathrm{E}[X_S] = (n/m)\cdot \mathrm{E}\left[\sum_{i=1}^{R} X_{a_i}\right] = (n/m)\cdot \sum_{i=1}^{R} \mathrm{E}[X_{a_i}] = (n/m)\cdot R\cdot (m/n) = R.$$

□

Exercise 6 *For a stream Σ of n items $x_1, x_2, \ldots, x_n$, consider the following algorithm:*

- *Run in parallel the Boyer-Moore majority algorithm and the Sticky Sampling algorithm with $\varphi = 1/2$.*
- *Let cand be the candidate identified by the Boyer-Moore algorithm, and S the set of items returned by the Sticky Sampling algorithm.*

If *cand* $\in S$ **then return** *cand* **otherwise return** *null.*

What can we say about the output of the algorithm? Distinguish among the case where Σ contains a majority element and the case where Σ does not contain a majority element.

Solution. Let cand be the item returned by the Boyer-Moore algorithm, and S the set returned by the Sticky Sampling algorithm.

- If a majority item A exists in the stream, then we know that cand $= A$, but the proposed algorithm will return it in output only if it is in S, which happens with probability at least $1 - \delta$, where δ is the probability parameter used in Sticky Sampling. Thus, the majority, if one exists, is returned with probability at least $1 - \delta$.
- If a majority element does not exist in Σ, then the algorithm returns null or an element in S. However, Sticky Sampling guarantees that any element in S has frequency at least $\varphi - \epsilon$, that is, $(1/2) - \epsilon$, since we set $\varphi = 1/2$. Then no element with frequency lower than $(1/2) - \epsilon$ is ever returned.

As an aside, we observe that the algorithm needs only in 1 pass, and the working memory size and the processing time per element are dominated by those of Sticky Sampling: namely, the working memory size is $O\left(\log(1/(\varphi\delta))/\epsilon\right) = O\left(\log(2/\delta)/\epsilon\right)$, and the processing time per element is $O\left(1\right)$ in expectation (using a Hash Table to implement the map). □

Exercise 7 (High Probability Guarantees (median trick)). *Consider a stream Σ with elements from a universe U, and let F_0 be the number of distinct elements in Σ. Suppose that you run ℓ* independent instances *of the probabilistic counting algorithm, obtaining ℓ estimates $\tilde{F}_0^{(j)}$ for F_0, with $1 \leq j \leq \ell$. Assume ℓ odd and let $\tilde{F}_0$ be the mediam value of the $\tilde{F}_0^{(j)}$'s. Show that with a suitable choice of $\ell = \Theta(\log |U|)$ we have:*

$$\Pr(\tilde{F}_0 < F_0/16) \leq 1/|U| \quad \textit{and} \quad \Pr(\tilde{F}_0 > 16F_0) \leq 1/|U|.$$

Hint: *Use the analysis of the Probabilistic counting algorithm and the Chernoff bound.*

Solution. Let us first show that $\Pr(\tilde{F}_0 < F_0/16) \leq 1/|U|$. Let m be the number of indices $j \in [1, \ell]$ such that $\tilde{F}_0^{(j)} < F_0/16$. It is immediate to see that $\tilde{F}_0 < F_0/16$ *if-and-only-if* more than half of the $\tilde{F}_0^{(j)}$'s are $< F_0/16$, hence *if-and-only-if* $m \geq (\ell+1)/2$. For every $j \in [1, \ell]$, let X_j be a Bernoulli random variable which is 1 if $\tilde{F}_0^{(j)} < F_0/16$, and 0 otherwise. By the analysis of the Probabilistic counting algorithm, we know that

$$p \stackrel{\text{def}}{=} \Pr(X_j = 1) = \Pr(\tilde{F}_0^{(j)} < F_0/16) \leq 1/16.$$

Since the X_j's are independent and $m = \sum_{j=1}^{\ell} X_j$, we have that m is a Binomial random variable with parameters ℓ and p, hence its expectation is $\mu = \ell \cdot p \leq \ell/16$. Observe that since $\mu \leq \ell/16$, we have that $(\ell+1)/2 \geq 6\mu$. Hence, by the above discussion and by applying the Chernoff bound we get that

$$\Pr(\tilde{F}_0 < F_0/16) = \Pr(m \geq (\ell+1)/2) \leq 2^{-(\ell+1)/2}.$$

By choosing ℓ such that $(\ell+1)/2 = \log_2 |U|$ (i.e., $\ell \in \Theta(\log |U|)$) we obtain $\Pr(\tilde{F}_0 < F_0/16) \leq 1/|U|$, as required.

The proof that $\Pr(\tilde{F}_0 > 16F_0) \leq 1/|U|$ is symmetrical. Let m' be the number of indices $j \in [1, \ell]$ such that $\tilde{F}_0^{(j)} > 16F_0$. By reasoning as above, we can show that

- $\tilde{F}_0 > 16F_0$ *if-and-only-if* $m' \geq (\ell+1)/2$; and
- m' is a Binomial random variable with parameters ℓ and p', and, from the analysis of the Probabilistic counting algorithm, we known that $p' = \Pr(\tilde{F}_0^{(j)} > 16F_0) \leq 1/16$.

Since the expectation of m' is $\mu' = \ell \cdot p' \leq \ell/16$ and $(\ell+1)/2 \geq 6\mu'$, by applying the Chernoff bound we get that

$$\Pr(\tilde{F}_0 > 16F_0) = \Pr(m' \geq (\ell+1)/2) \leq 2^{-(\ell+1)/2}.$$

By choosing ℓ such that $(\ell+1)/2 = \log_2 |U|$ (i.e., $\ell \in \Theta(\log |U|)$) we obtain $\Pr(\tilde{F}_0 > 16F_0) \leq 1/|U|$, as required.

Finally, as an aside, we observe that the working memory requirements and the update and query time of the improved estimation algorithm are ℓ times larger than those of the standard probabilistic counting algorithm. $\square$

Exercise 8 *Consider a $d \times w$ count sketch C for a stream Σ of items from a universe U, and let $\tilde{F}_{2.j}$ the estimate of the second moment F_2 obtained from row j of C, namely*

$$\tilde{F}_{2.j} = \sum_{k=0}^{w-1} (C[j,k])^2$$

for $0 \le j < d$. Show that $\tilde{F}_{2.j}$ is an unbuiased estimate of F_2, that is, $\mathrm{E}[\tilde{F}_{2.j}] = F_2$.

Solution. For $0 \le j < d$, let h_j and g_j be the two hash functions associated with row j of C. Also, for every item $a \in U$, let f_a be the absolute frequency of a in Σ. For every pair of items $a, b \in U$ with $a \neq b$ we define the following random variable:

$$Y_{ab} = \begin{cases} f_a f_b & \text{if } h_j(a) = h_j(b) \wedge g_j(a) = g_j(b) \\ -f_a f_b & \text{if } h_j(a) = h_j(b) \wedge g_j(a) = -g_j(b) \\ 0 & \text{else} \end{cases}$$

Note that $Y_{ab} = f_a f_b$ with probability $1/(2w)$ and, likewise, $Y_{ab} = -f_a f_b$ with probability $1/(2w)$, while $Y_{ab} = 0$ with probability $1 - 1/w$. Therefore, the expectation $\mathrm{E}[Y_{ab}]$ is 0. Now, consider an arbitrary cell $C[j,k]$ and let $X \subseteq U$ be the items mapped by h_j to this cell. Then, it is easy to see that

$$(C[j,k])^2 = \sum_{a \in X} (f_a)^2 + \sum_{a,b\, \in X,\, a \neq b} 2 f_{ab}.$$

From this fact, we easily derive that

$$\tilde{F}_{2,j} = \sum_{a \in U} (f_a)^2 + \sum_{a,b\, \in U,\, a \neq b} 2 f_{ab} = F_2 + \sum_{a,b\, \in U,\, a \neq b} 2 f_{ab}$$

Therefore,

$$\mathrm{E}[\tilde{F}_{2,j}] = F_2 + \mathrm{E}\left[\sum_{a,b\, \in U,\, a \neq b} 2 f_{ab} \right] = F_2 + \sum_{a,b\, \in U,\, a \neq b} 2\mathrm{E}[f_{ab}] = F_2,$$

where in the last passage we used the fact the $\mathrm{E}[f_{ab}] = 0$, which was observed before. $\square$

Exercise 9 *Consider a stream $\Sigma = x_1, x_2, \ldots, x_n$ of n measurements from sensors. Each measurement x_i is a pair (k_i, w_i), where k_i is the ID of a sensor and w_i is the value of the measurement (an integer). For a given sensor u occurring in Σ define*

$$f_u = \sum_{(k_i, w_i) \in \Sigma \,:\, k_i = u} w_i,$$

i.e., the aggregate measurements taken by u.

1. *Briefly describe a space-efficient unbiased estimator for* $H_2 = \sum_u (f_u)^2$*, where the sum is over all sensors occurring in the stream.*
2. *What can you say about the unbiasedness of your estimator?*

Solution.

1. The estimator is a simple variant of the count sketch. Let U be the universe to which the IDs of the sensors belong. The sketch uses the following ingredients:

   - a $d \times w$ array C of counters, all initialized to 0;
   - d hash functions $h_j : U \to \{1, 2, \ldots, w\}$, for $1 \le j \le d$;
   - d hash functions $g_j : U \to \{-1, +1\}$, for $1 \le j \le d$.

   When the element $x_t = (k_t, w_t)$ of the stream Σ arrives, the array C is modified as follows. For every $1 \le j \le d$:

   $$C[j, h_j(k_t)] \leftarrow C[j, h_j(k_t)] + g_j(k_t) \cdot w_t.$$

   At the end of the stream, we define

   $$\tilde{H}_{2,j} = \sum_{\ell=1}^{w} (C[j, \ell])^2,$$

   for every $1 \le j \le d$, and define $\tilde{H}_2$ to be the median of the $\tilde{H}_{2,j}$'s. We consider $\tilde{H}_2$ as estimate of the *true* value $H_2 = \sum_u (f_u)^2$.

2. Consider a stream Σ' of sensor IDs (i.e., elements of U) obtained from Σ by substituting each pair (k_i, w_i) with w_i occurrences of k_i. In order to distinguish betwen Σ and Σ', we denote frequencies in the two streams as follows. For every sensor u define

   $$\begin{aligned} f_u^{\Sigma} &= \sum_{(k_i, w_i) \in \Sigma \,:\, k_i = u} w_i \\ f_u^{\Sigma'} &= \sum_{k_i \in \Sigma' \,:\, k_i = u} 1. \end{aligned}$$

   Clearly, $f_u^{\Sigma} = f_u^{\Sigma'}$, for every u. Hence, by letting F_2 denote the second moment of Σ' (i.e., $F_2 = \sum_u (f_u^{\Sigma'})^2$) we have that $F_2 = H_2$. Moreover, it is immedate to see that at the end of the stream Σ, the count sketch C is the same as the one that we would obtain for Σ'. By what we learned in class, we know that $\mathrm{E}[\tilde{H}_{2,j}] = \mathrm{E}[\sum_{\ell=1}^{w} (C[j, \ell])^2] = F_2$, for every $1 \le j \le d$. Since $F_2 = H_2$, we conclude that $\mathrm{E}[\tilde{H}_2] = H_2$, which shows that the estimator is unbiased.

$\square$

Exercise 10 *Let $\Sigma = x_1, x_2, \ldots, x_n$ be a stream of $n > 10$ items (n even), which contains $(n/2) - 5$ occurrences of item a, $(n/2) - 5$ occurrences of item b, and 10 occurrences of other items $\neq a, b$. Let C be a* **Count-min sketch** *of size $d \times w$ for Σ. We define the following notation for item a:*

$$\begin{aligned} f_a &= \text{true frequency of } a \text{ in } \Sigma; \\ \tilde{f}_{a,j} &= \text{the estimate of } f_a \text{ provided by row } j \text{ of } C, \text{ with } 0 \le j < d; \\ \tilde{f}_a &= \text{the final estimate of } f_a \text{ provided by the sketch } C. \end{aligned}$$

Answer to the following points, **assuming** $w = 2$.

1. *Suppose that in row j, a and b are mapped to a different counter. Give a tight upper bound R to the relative error $(\tilde{f}_{a,j} - f_a)/f_a$.*

2. *Inspired by the previous point, compute the following probabilities*
   - $\Pr\left((\tilde{f}_{a,j} - f_a)/f_a) \le R\right)$, *for* any given row j, *with* $0 \le j < d$.
   - $\Pr\left((\tilde{f}_a - f_a)/f_a) \le R\right)$.

Solution.

1. If a and b are mapped to a different counter of row j, we must have
$$f_a \le \tilde{f}_{a,j} \le f_a + 10.$$
Since $f_a = (n/2) - 5$, we have that
$$\frac{\tilde{f}_{a,j} - f_a}{f_a} \le \frac{10}{(n/2) - 5}.$$
Therefore, a good upper bound is $R = 10/((n/2) - 5)$.

2. We compute the required probabilities as follows.
   - From the previous point, it is easy to see that the probability that $((\tilde{f}_{a,j} - f_a)/f_a) \le R$ is (at least) the probability that a and b are mapped to different counters, which is 1/2. Thus, for any given row j, with $0 \le j < d$, we have
   $$\Pr\left((\tilde{f}_{a,j} - f_a)/f_a) \le R\right) \ge \frac{1}{2},$$
   (The probability is exactly 1/2 when n is large.)
   - Recall that the estimate for f_a provided by the Count-min sketch C is $\tilde{f}_a = \min_{0 \le j < d} \tilde{f}_{a,j}$. Thus, $(\tilde{f}_a - f_a)/f_a = \min_{0 \le j < d} (\tilde{f}_{a,j} - f_a)/f_a$. This immediately implies that $(\tilde{f}_a - f_a)/f_a > R$ only if $(\tilde{f}_{a,j} - f_a)/f_a > R$ for all j's, which, based on the previous point, happens with probability $\le (1/2)^d$. Hence,
   $$\Pr\left((\tilde{f}_a - f_a)/f_a) \le R\right) \ge 1 - \frac{1}{2^d}.$$
   (Again, the probability is exactly $1/2^d$ when n is large.)

□

Exercise 11 *Consider a Bloom filter built to assess membership for a set S of m elements. The Bloom filter consists of an n-bit array A and k hash functions $h_0, h_1, \ldots, h_{k-1}$, mutually independent and with values uniformly distributed in $[0, n-1]$. Assume that n is even, and that each hash function h_i is such that, for every $e \in S$, $h_i(e) \bmod n/2$ is uniformly distributed in $[0, n/2-1]$.*

1. *Show how to transform, in $O(n)$ time, the given Bloom filter into a new Bloom filter based on an $n/2$-bit array B, and describe how to assess membership with the new Bloom filter.*

2. *Compute the probability that a given cell $B[i]$ of the new array is 0.*

Solution.

1. We define the new array B as follows:

$$B[i] = A[i] \text{ OR } A[i + n/2],$$

for every $0 \leq i < n/2$. It is immediate to see that the array B is *exactly* the same as the one that would be constructed using the k hash functions h'_j, for $1 \leq j \leq k$, where for each $e \in S$, $h'_j(e) = h_j(e) \bmod n/2$. An element x is claimed to be in S if $h'_j(x) = 1$ for each $1 \leq j \leq k$. Note that the construction of B from A can be accomplished in $O(n)$ time without using the set S.

2. Since we assumed that for every $e \in S$ and $0 \leq j < k$, $h_j(e) \bmod n/2$ is uniformly distributed in $[0, n/2-1]$, the hash functions h'_j's satisfy the hypotheses of independence and uniform distribution made in the analysis of the Bloom filter presented in class. By repeating that analysis, we can conclude that, for sufficiently large n, the probability that any given cell $B[i]$ is 0, is very close to $e^{-km/(n/2)} = e^{-2km/n}$.

□